LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 9: Project: Smart Assistant App

Module 3 | 50 minutes | Project: input, variables, decisions | Grades 6–8 · ages 11–14

Objective: Plan, build and test a helpful assistant app that asks questions, stores answers in variables, calculates, and responds with decisions.

Success criteria

- My app follows input → process → output.
- It calculates and decides.
- I tested it with a test table, including invalid input.

Key vocabulary

- **validation**: Checking input is sensible.
- **test table**: A plan of inputs and expected outputs.
- **IPO**: Input → Process → Output.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Human app (5 min): one learner is the app, following an IPO card; others give inputs.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Input, process or output?. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Smart Assistant.

Unplugged warm-up

Human app (5 min): one learner is the app, following an IPO card; others give inputs.

Teaching the idea

The project integrates input, variables, operators, selection and output, plus input validation and test tables (normal/boundary/invalid).

Analogy: A vending machine: takes input, checks it, processes it, gives output.

Live demo: Break a finished assistant with a silly input (“banana”, -5); then add validation.

Questions to ask

Ask	Listen for
What are your inputs?	The questions asked.
Where is the decision?	The if/else block.
Which input broke your app?	Test-table discussion.

Differentiation

Support
Provide the study-planner starter; learners customise it.

Stretch
Remember the user’s name and use it in every reply.

Common misconceptions

Misconception	How to fix it
Users always type numbers	Validate input.
Testing once is enough	Test normal, boundary and invalid cases.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Input & variables	None	Asks but doesn’t store	Stores inputs in named variables	Clear names; reuses values well
Processing	None	Calculation or decision	Calculation and decision	Chained decisions and Boolean logic
Output	None	Fixed messages	Joined, personalised replies	Helpful, friendly and clear
Validation	None	Mentioned	Handles one invalid case	Handles several invalid cases kindly
Testing	None	A few tries	Test table with normal and boundary tests	Thorough table; bugs found and fixed

Exit ticket
One invalid input your app now handles.

Homework
Ask someone at home to use your app and add their inputs to your test table.

Answer key — Kitabash activities

1. Input, process or output? (perform · categorize)

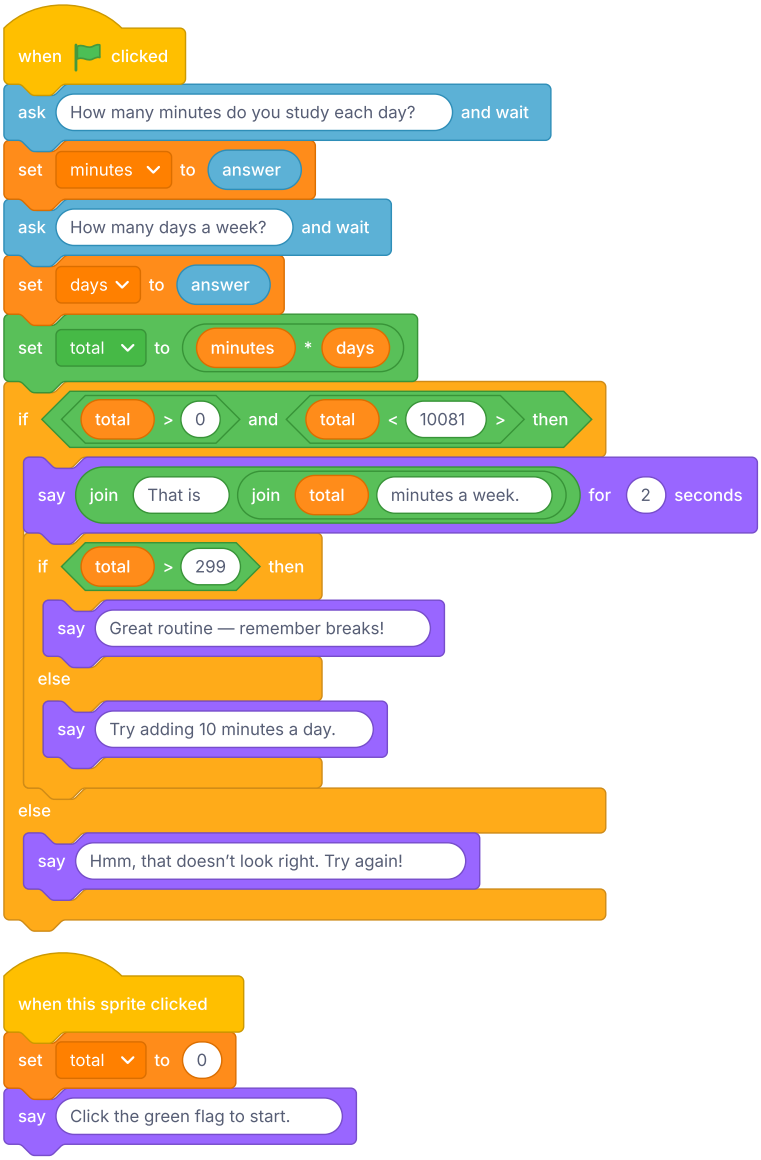
Input: ask [How many minutes?] and wait, set name to answer
Process: set total to minutes × 7, if total > 300 then …
Output: say (join [Weekly:] (total))

2. Project: Smart Assistant (probe · code lab)

Goals checked: Asks 2+ questions · Stores answers · Calculates · Decides · Builds a reply

Example project:

Bot



Answer key — worksheet

1. **Key words:** validation = Checking input is sensible. · test table = A plan of inputs and expected outputs. · IPO = Input → Process → Output.
2. **Input, process or output?:** Input: ask [How many minutes?] and wait, set name to answer · Process: set total to minutes × 7, if total > 300 then … · Output: say (join [Weekly:] (total))
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.